

The Muscular System

OUR MUSCLES:

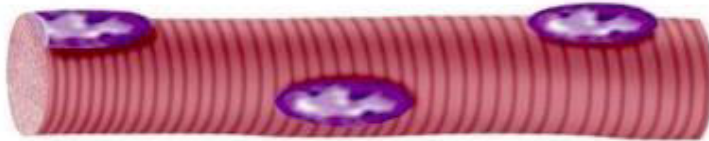
- Enable us to move our body parts.
- Give us our individual shape.
- Protect and keep in place our abdominal organs.
- Enable us to maintain good posture.
- Help in the circulation of our blood.
- Generate body heat when they contract.

THERE ARE OVER 639 SKELETAL MUSCLES
IN THE BODY- 150 IN THE HEAD AND NECK.

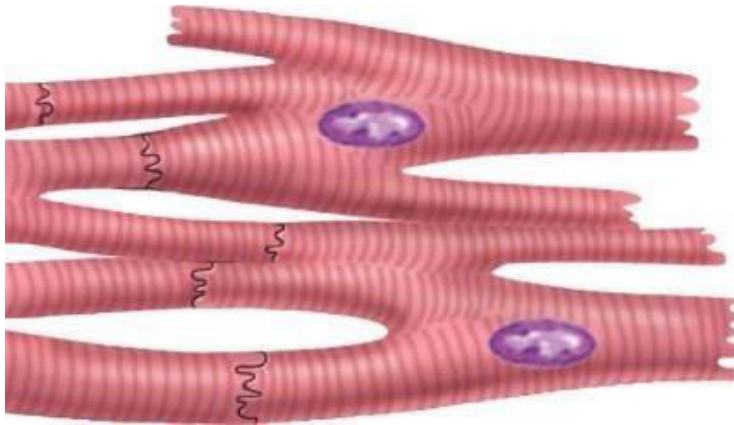
Functions of Muscular Tissue

- Like nervous tissue, muscles are excitable or "irritable"
 - they have the ability to respond to a stimulus
- Unlike nerves, however, muscles are also:
 - **Contractible** (they can shorten in length)
 - **Extensible** (they can extend or stretch)
 - **Elastic** (they can return to their original shape)

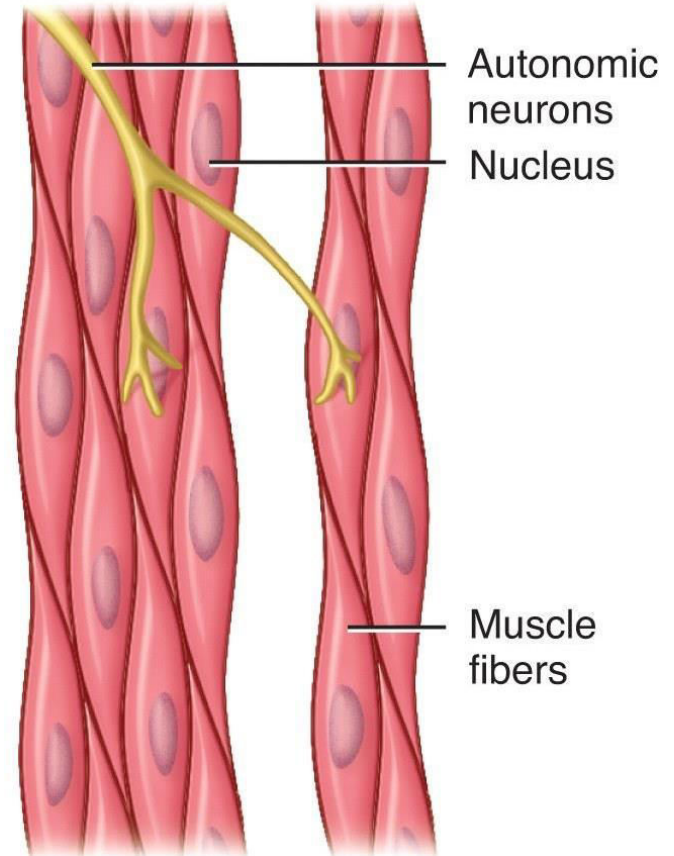
Three Types of Muscular Tissue



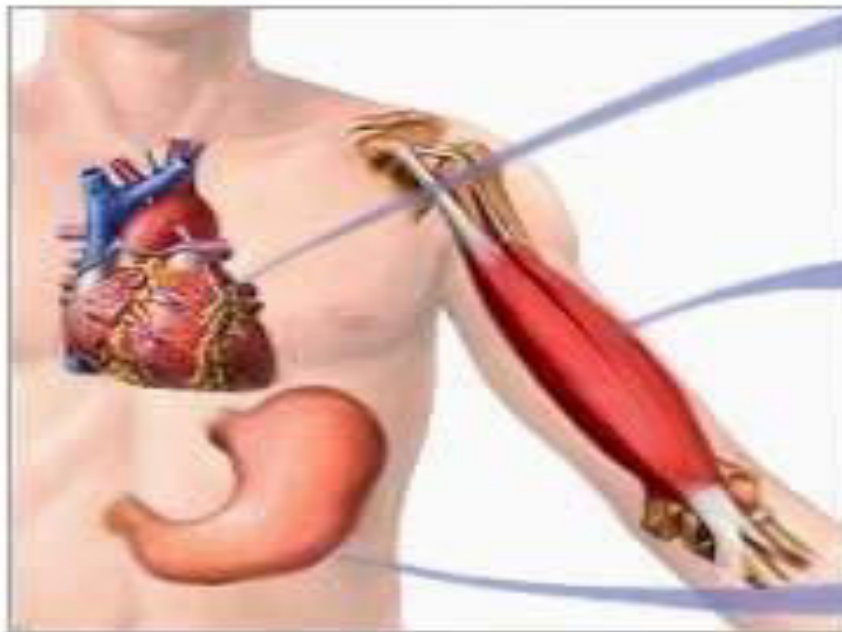
(a) Skeletal muscle



(b) Cardiac muscle



(c) Visceral smooth muscle



Cardiac muscle cell

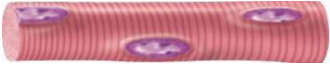
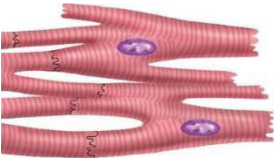


Skeletal muscle cell



Smooth muscle cell

Three Types of Muscular Tissue

	Location	Function	Appearance	Control
Skeletal 	skeleton	movement, heat, posture	striated , multi-nucleated (eccentric), fibers parallel	voluntary
Cardiac 	heart	pump blood continuousl y	striated , one central nucleus	involuntary
Visceral (smooth muscle) 	G.I. tract, uterus, eye, blood vessels	Peristalsis, blood pressure, pupil size, erects hairs	no striations , one central nucleus	involuntary

Muscular System

- **Skeletal muscle is the only organ of the muscular system**
- **Skeletal muscle is composed of skeletal muscle tissue and also contains nervous tissue, blood vessels and connective tissue**
- **Half of the body's weight is muscle tissue**
 - **Skeletal muscle = 40% in males, 32% in females**
 - **Cardiac muscle = 10%**

Skeletal muscle fibers (cells) are arranged into bundles called fascicles

Fascicles are bound by connective tissue

Four different connective tissue coverings

Deep fascia

Surrounds entire skeletal muscle and extends beyond its length

Epimysium

Closely surrounds skeletal muscle, binds fascicles together

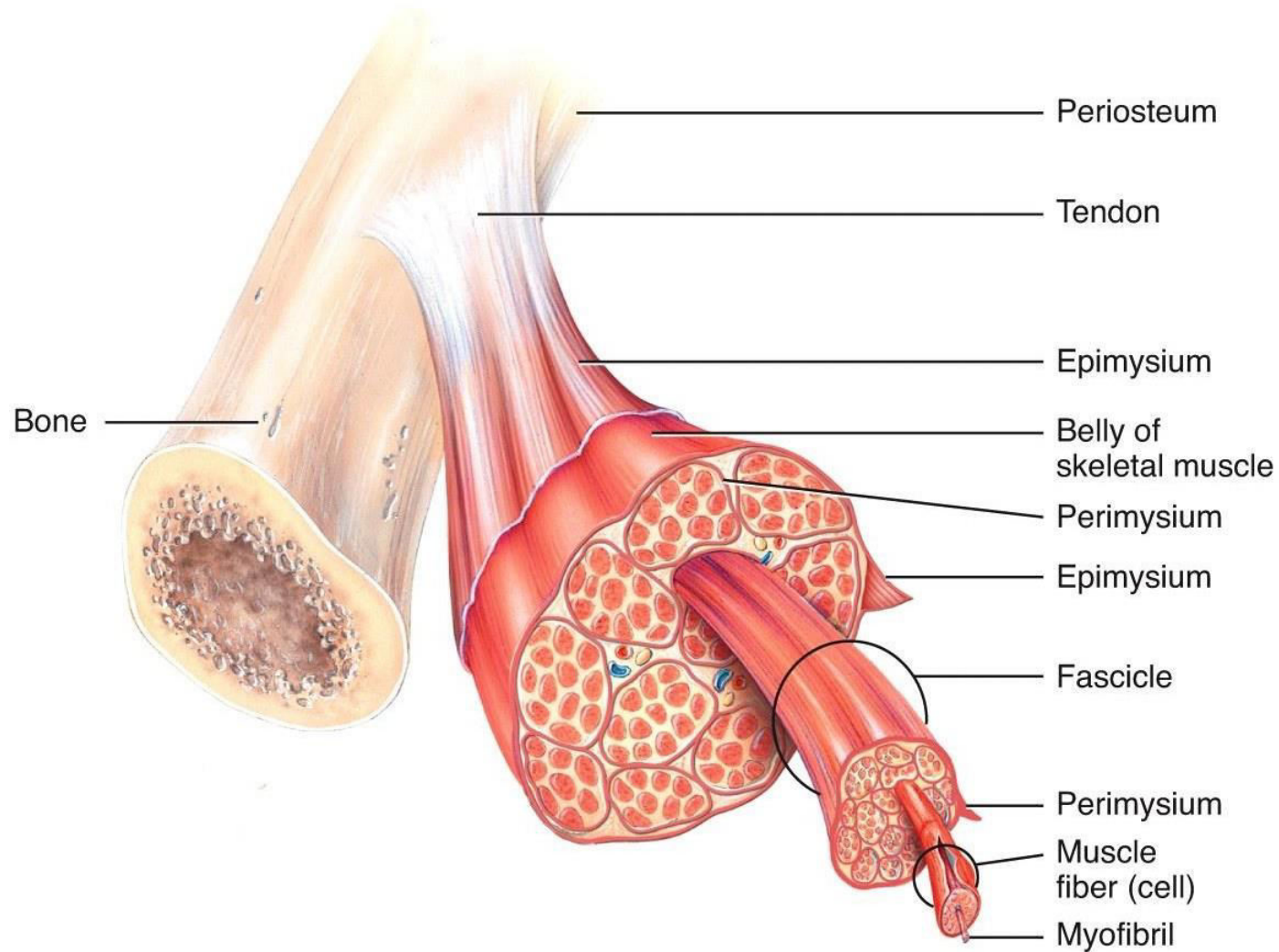
Perimysium

Surrounds each fascicle

Endomysium

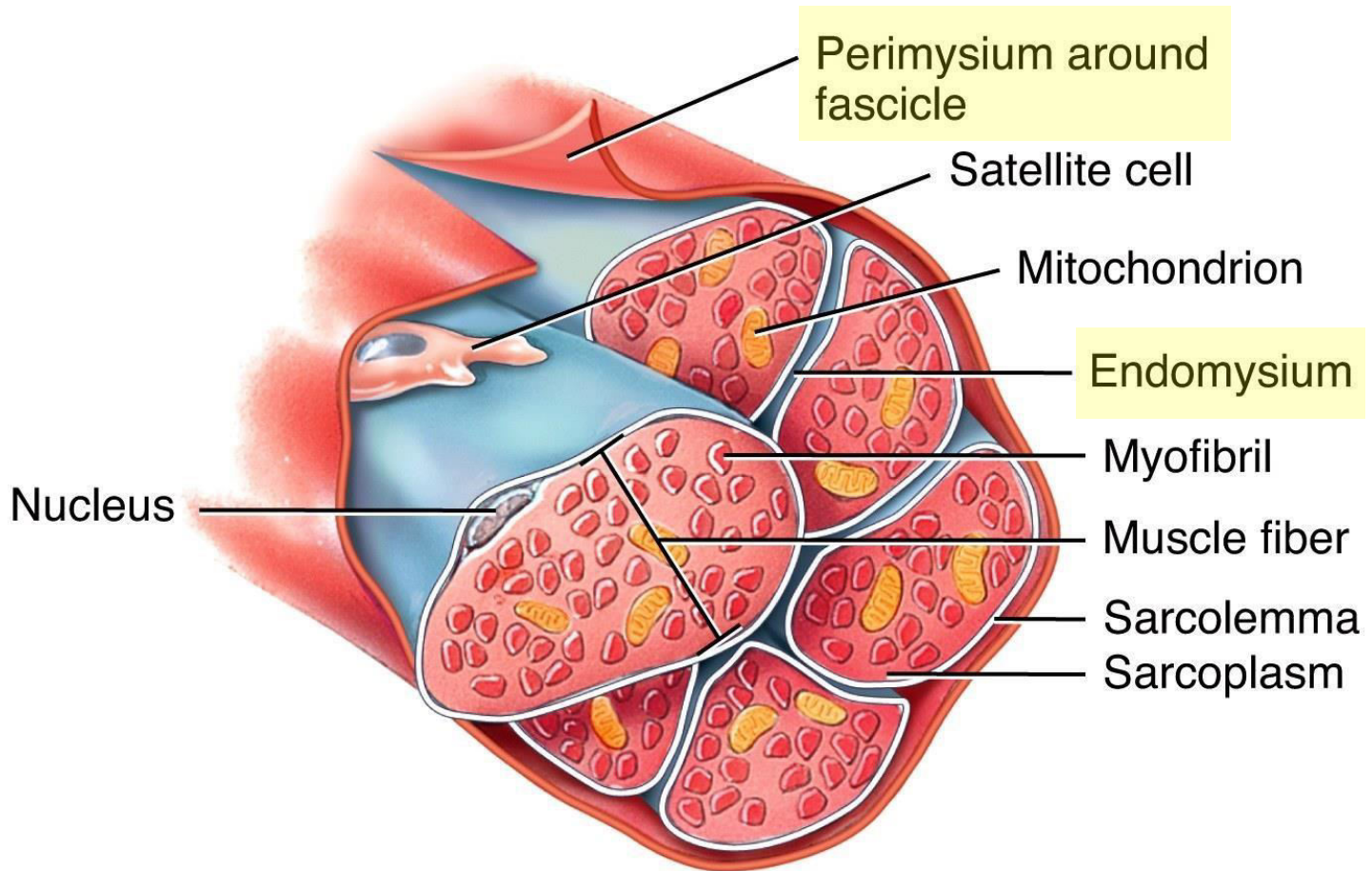
Surrounds each muscle fiber (cell)

Organization of Muscle Tissue



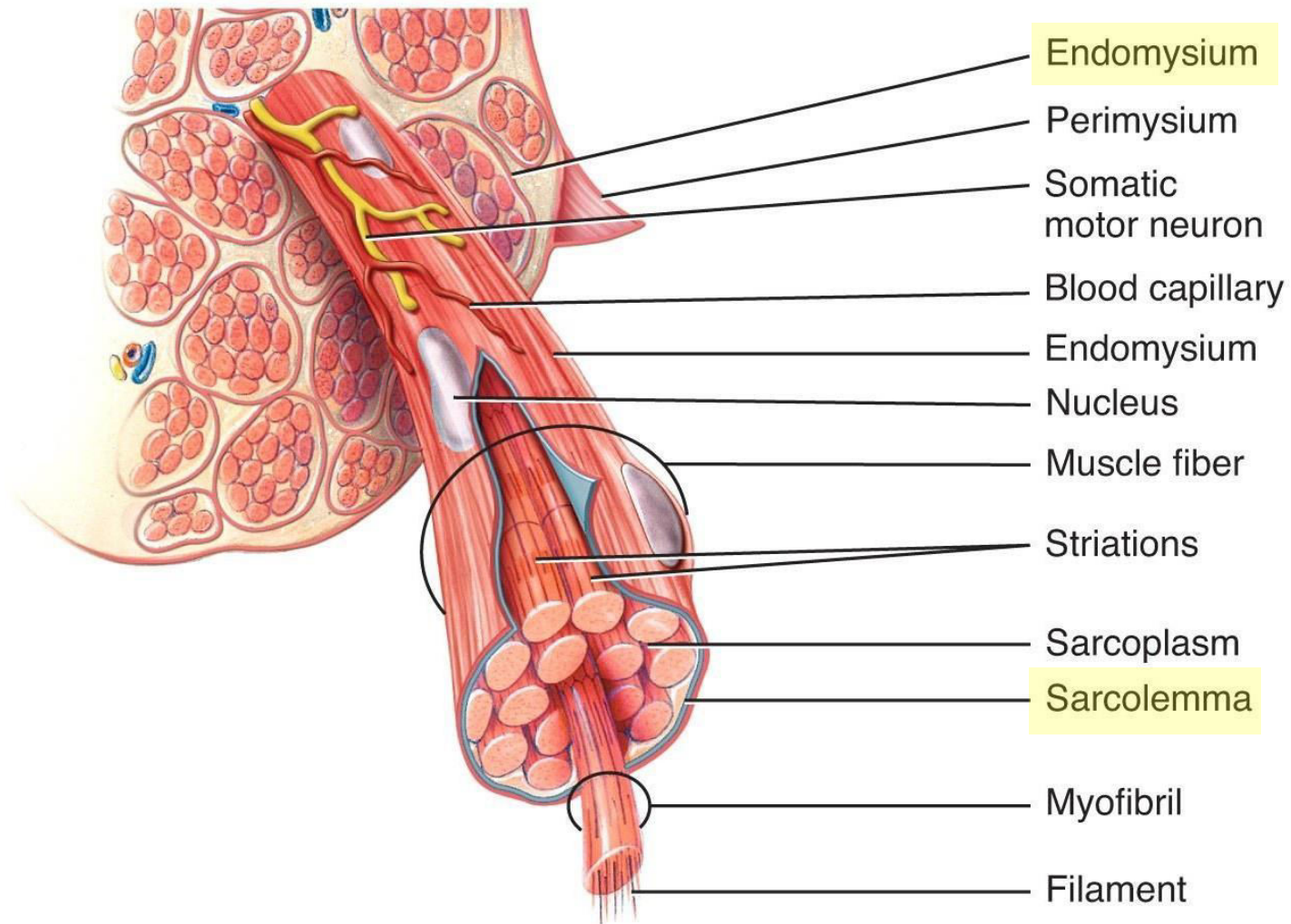
A muscle, a fasciculus, and a fiber all visualized

Organization of Muscle Tissue



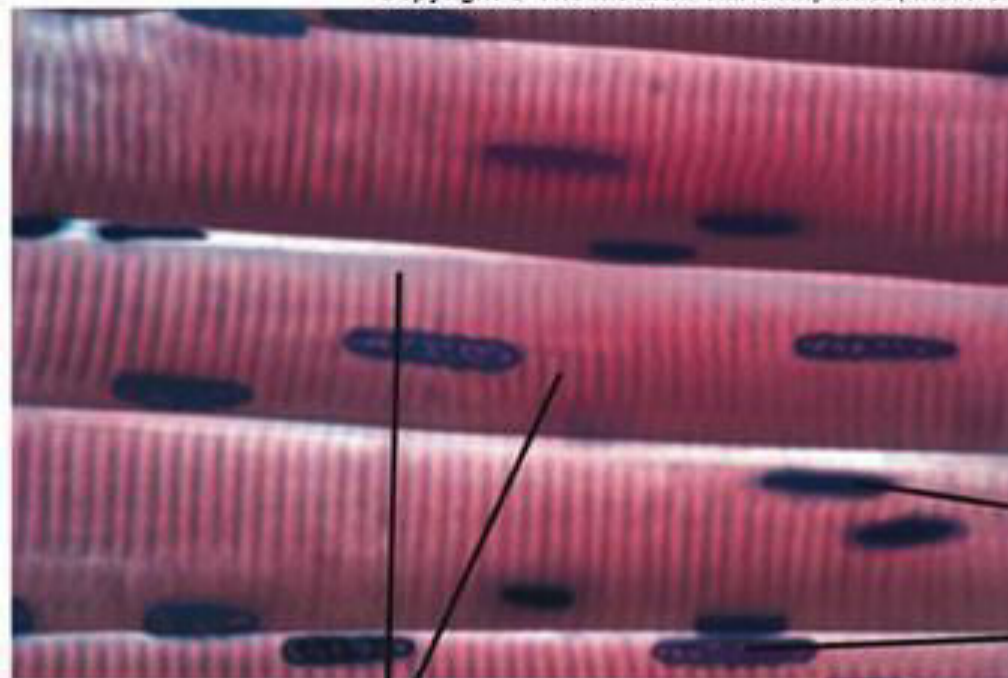
Organization of a fasciculus

Organization of Muscle Tissue



Organization of a muscle fiber

Skeletal Muscle Tissue



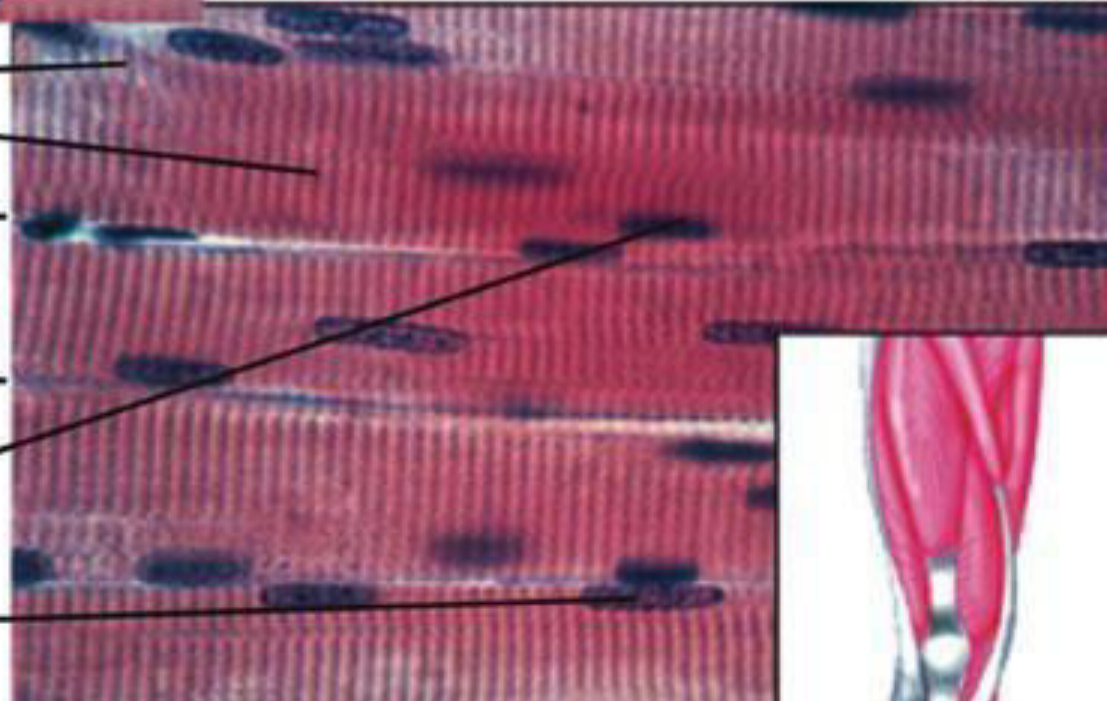
Muscle fiber

Nuclei

Striations

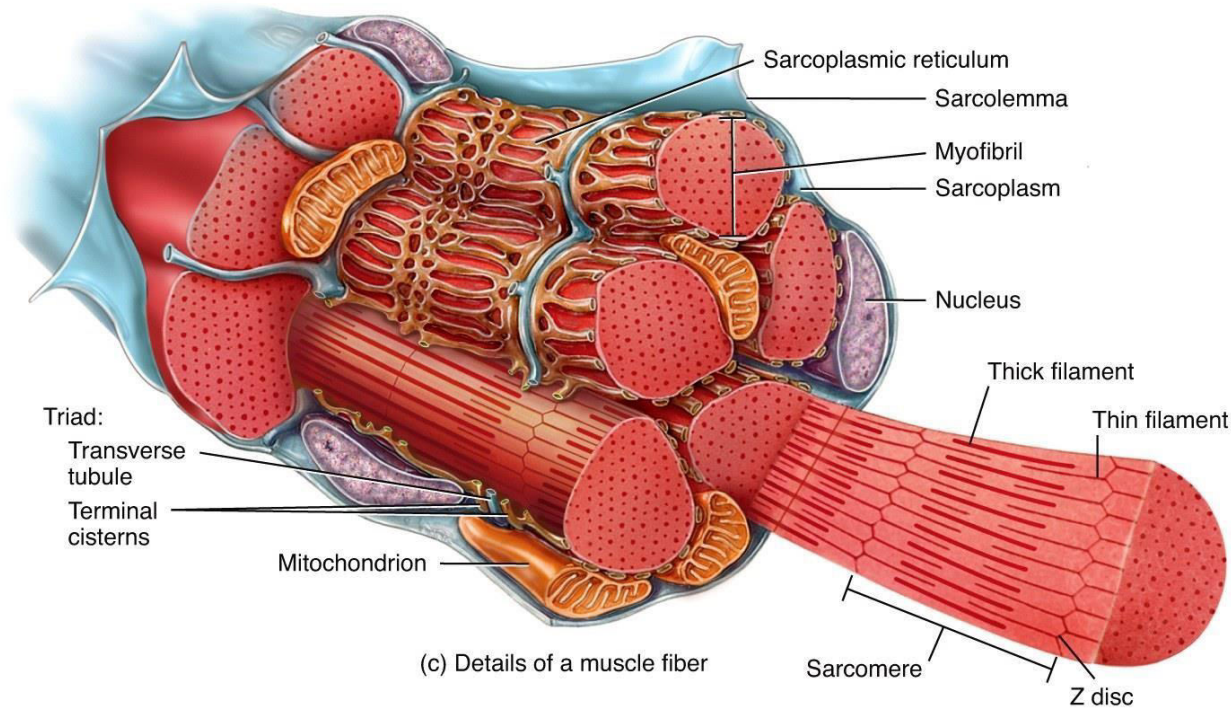
Muscle fiber

Nuclei



Skeletal Muscle Arrangement

- A single muscle cell is a muscle fiber
 - Fibers are made up of myofibrils
 - Myofibrils are made up of thick and thin filaments
- **Sarcolemma** – muscle cell membrane
- **Sarcoplasm**
- muscle cell cytoplasm



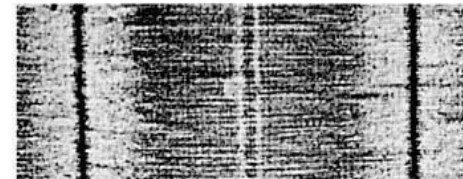
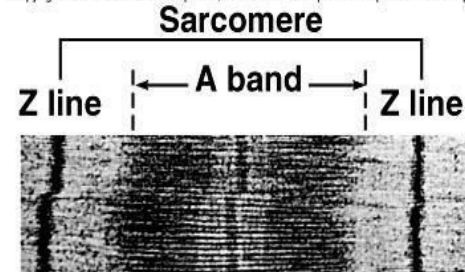
Sliding Filament Theory

- A sarcomere is the functional unit of skeletal muscle
- When a skeletal muscle contracts, sarcomeres shorten
- This is described by the sliding filament theory

Contraction in the Sarcomere

- A band stays the same
- I band gets smaller
- H zone gets smaller
- Sarcomere shortens

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(b)

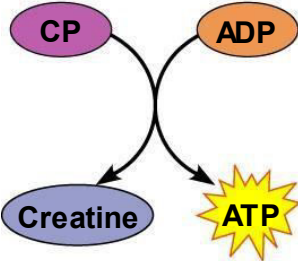
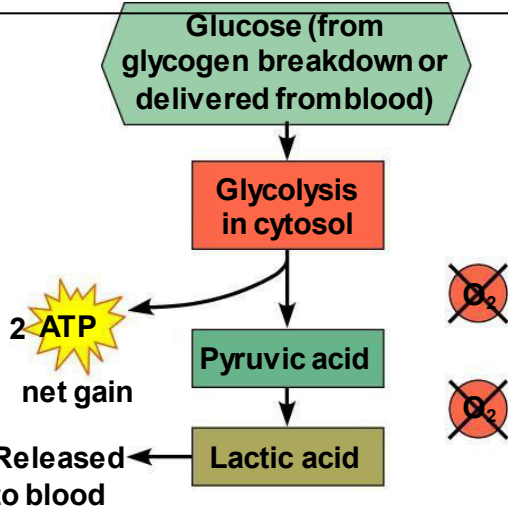
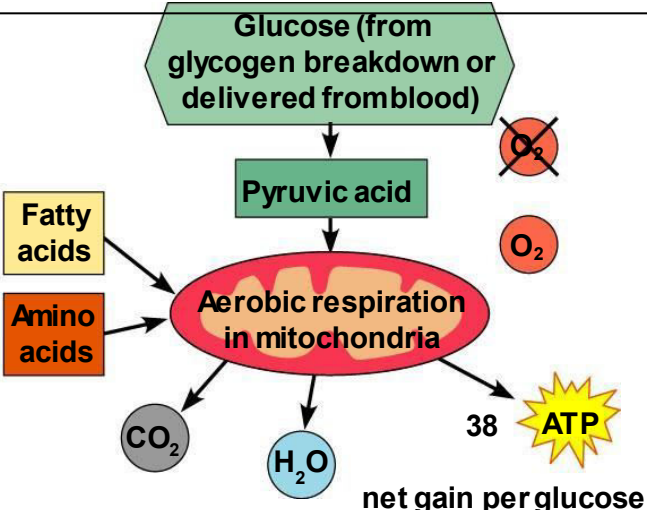
Energy for Contraction

- Muscle cells require huge amounts of ATP energy to power contraction
- The cells have only a very small store of ATP
- *Three pathways supply ATP to power muscle contraction*

Energy for Contraction

- ATP initially supplied from cellular respiration
- If ATP is abundant, is converted to creatine phosphate and stored in skeletal muscles
- When ATP is low, creatine phosphate supplies phosphate to ADP making ATP
- CP & ATP stores only good for about a 10 second maximal contraction
- ATP must then come from cellular respiration or glycolysis

Methods of regenerating ATP during muscle activity

		
(a) Direct phosphorylation [coupled reaction of creatine phosphate (CP) and ADP]	(b) Anaerobic mechanism (glycolysis and lactic acid formation)	(c) Aerobic mechanism (aerobic cellular respiration)
Energy source: CP	Energy source: glucose	Energy source: glucose; pyruvic acid; free fatty acids from adipose tissue; amino acids from protein catabolism
Oxygen use: None Products: 1 ATP per CP, creatine Duration of energy provision: 15s	Oxygen use: None Products: 2 ATP per glucose, lactic acid Duration of energy provision: 30–60 s.	Oxygen use: Required Products: 38 ATP per glucose, CO₂, H₂O Duration of energy provision: Hours

Oxygen & Muscle Contraction

- Myoglobin of muscle (similar to hemoglobin) binds to and stores oxygen
- Supplies O_2 needed to make ATP for contraction

Exercise and Skeletal Muscles

- Prolonged, moderate exercise
 - ATP supplied through cellular respiration
 - Once glycogen stores are depleted in muscle, glucose and fatty acid deliveries from blood are used as fuel source

Exercise and Contraction

- **In intense strenuous activity, oxygen can be depleted**
- **Why?**
- **Contraction of skeletal muscles decreases blood delivery to muscles**
- **Nutrient and O₂ levels in contracting muscles decrease**

Exercise and Skeletal Muscles

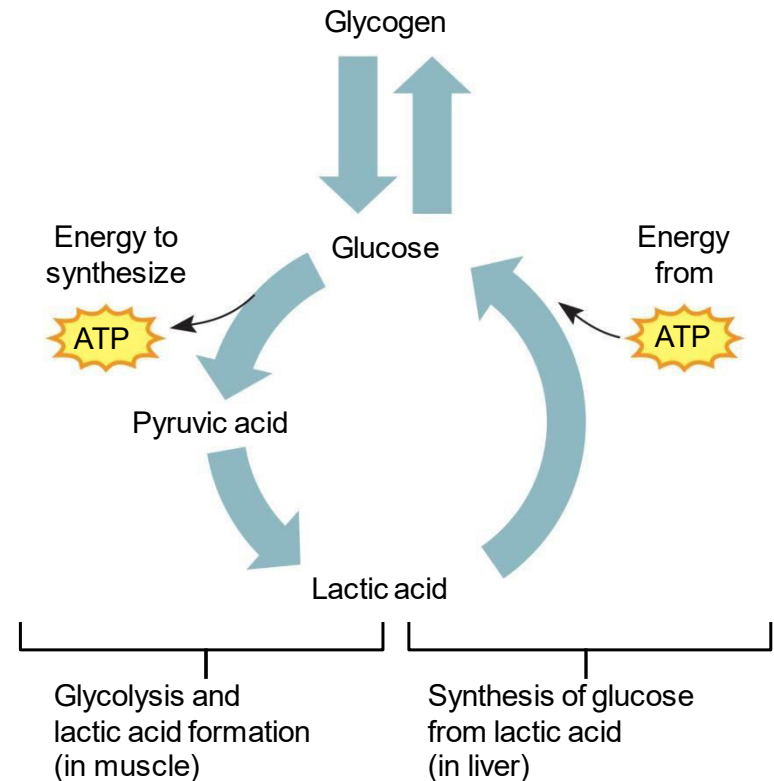
- Intense, strenuous exercise
 - Muscles exceed capacity of respiratory and cardiovascular systems to deliver oxygen for contraction
 - ATP supplied anaerobically through glycolysis
 - Pyruvate is converted to lactic acid
 - Lactic acid builds up in muscles
 - Causes muscles to fatigue

Oxygen Debt

- *Amount of oxygen needed by liver cells to use the accumulated lactic acid to produce glucose*

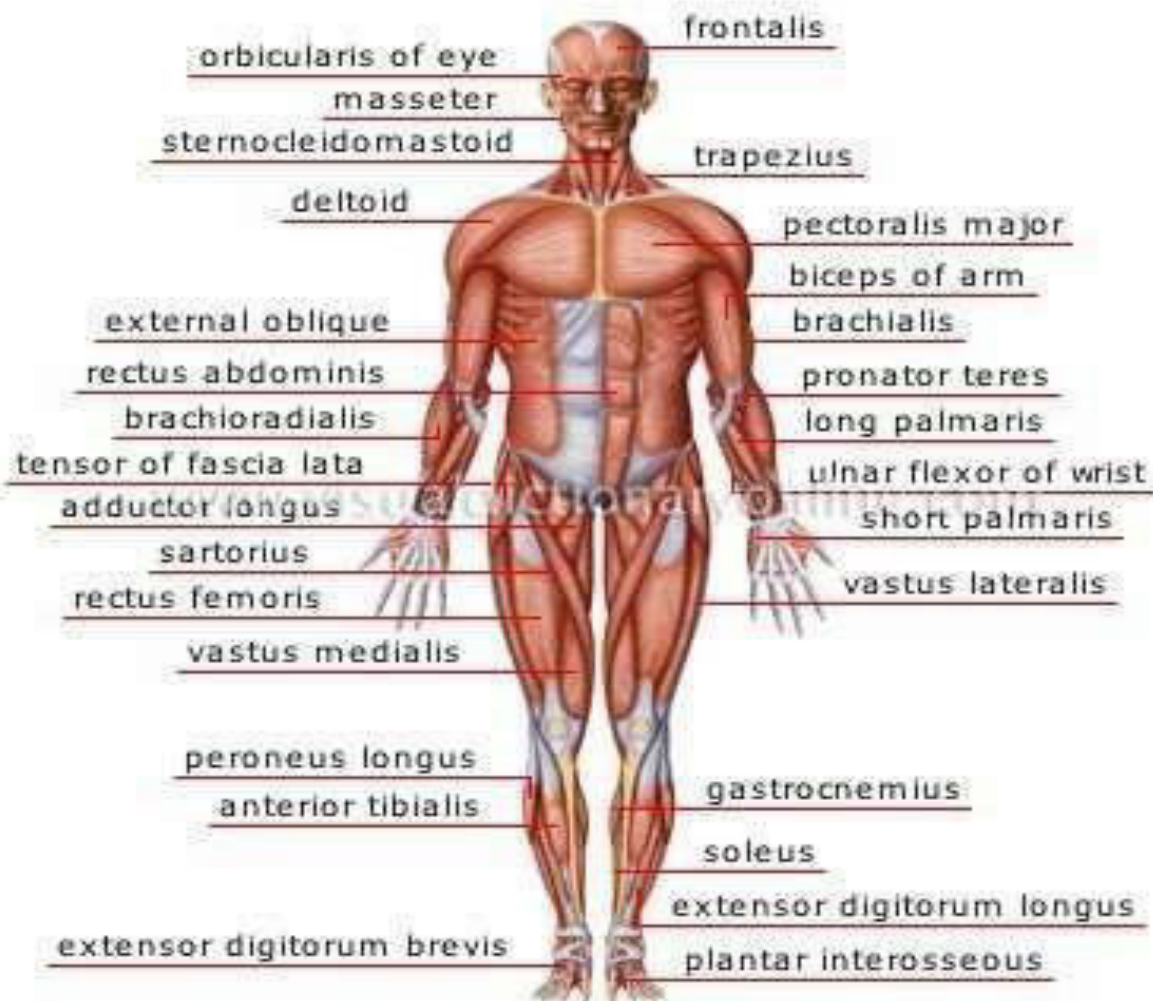
- Oxygen not available
- Glycolysis continues
- Pyruvic acid converted to lactic acid
- Liver converts lactic acid to glucose
- *Also the amount of oxygen needed to replace O_2 levels in skeletal muscle to pre-exercise levels*

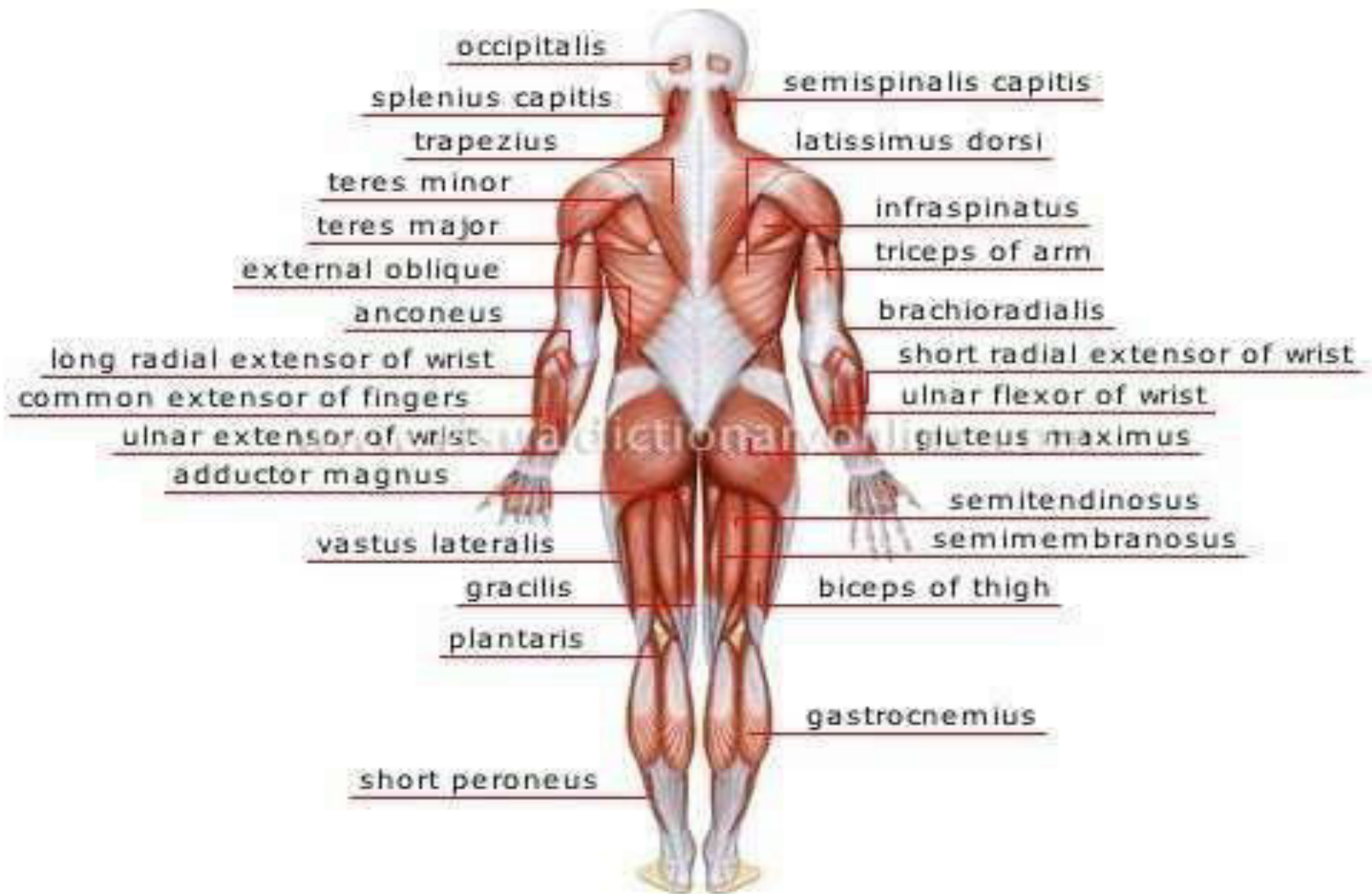
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Heat Production

- **Cellular respiration is only about 40% efficient**
- **About 60% of the energy found in a glucose is lost as heat during cellular respiration**
- **Muscle contraction generates heat because muscles use large amounts of nutrients to make ATP, generating large amounts of heat**
- **Heat is used to maintain body temperature**





How many muscles are present

Answer: 639 approx. Muscles

The muscles make up about 40 % of the body mass.

Q. What is the longest muscle in the body?

Answer: *The Sartorius*

The Sartorius runs from the outside of the hip, down and across to the inside of the knee. It twists and pulls the thigh outwards.

Q . What is the smallest muscle in the body?

Answer: *The Stapedius*

The Stapedius is located deep in the ear. It is only 5mm long and thinner than cotton thread. It is involved in hearing.

Q What is the biggest muscle in the body?

Answer: *The Gluteus Maximus*

The Gluteus Maximus is located in the buttock. It pulls the leg backwards powerfully for walking and running.

THANKS